OPS-AVI-REC-01

Recommended Electrical Components

Avionics Department

Table 1. Document History

Version	Comments	Date	Author	Approved by
V0.01.0	Created document	23/07/24	RH	-
V0.01.1	Updated formatting, added in Fuse and Diode information	27/10/24	RH	-
V0.01.2	Formatted document, removed some duplicate information	20/12/24	RH	-
V0.01.3	Fixed document formatting	15/06/25	RH	-
V1.00.0	Migrated to new formatting, updated some links to be shorter, added USB connectors, power connectors, more diodes and ICs Added table formats for components	18/2/26	RH, AH	KH

Table 2. Approval

Role	Name	Signature	Date
Avionics Lead	Raphael Ho	RH	12/3/2026
ADCS Lead	Aiden Hains	AH	16/3/2026
Systems Engineering Lead			
Supervisor / Advisor			

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Table 3. Related Documents

Document Number / Link	Name
Avionics Inventory.xlsm	Inventory

Table 4. Nomenclature / Acronyms

Acronym	Meaning
ESD	Electrostatic Discharge
FPC	Flexible Printed Circuit
IC	Integrated Circuit
MOSFET	Metal Oxide Semiconductor Field Effect Transistor
PAST	Perth Aerospace Student Team
PCB	Printed Circuit Board
PPM	Parts Per Million
SMT	Surface Mount Technology
TCXO	Temperature Compensated Crystal Oscillator
USB	Universal Serial Bus

Table 5. Roles and Responsibility

Role	Responsibility
Avionics Lead	Responsible for ensuring that this document is being updated
Avionics Lead, Avionics 2IC, ADCS Lead, ADCS 2IC	Responsible for ensuring that this document is being implemented
All PAST Members & Onboarding Members	Responsible for following the operations outlined in this document

Document Purpose

This document aims to outline strongly recommended components that should be used when designing a PCB. Issues can arise when unfamiliar components are used, which can cause issues such as ordering the wrong series or not using already available components in the PAST Inventory.

This document's audience is intended for anyone designing a PCB and requires selecting components.

Background

These recommended components have been compiled through already tested PCBs and some from the guidance of advisors.

Commonly used components such as Resistors, Capacitors, Connectors, and LEDs are **strongly recommended** to follow these recommendations. Any deviation for those components **must** be specified in the design documentation.

The commonly used components will **not need** to be procured as they are bulk purchased by Avionics every semester. However, if they are nearly used up, this must be notified to the Avionics department.

Resistors

Deviations are highly not recommended.

Table 6. Primary Preference Resistor

Primary Preference	
Yaego AC Series, 0603 size	
Datasheet	https://www.yageo.com/upload/media/product/products/datasheet/rchip/PYu-AC_E_105_RoHS_L_2.pdf
Resistance Range	• 10 Ω - 1MΩ • Jumper (0Ω) • Note: Maximum 7.5 MΩ for 1206 size
Part Number Specifications	AC 0603 [X_1] R [X_2] [XX_3] [$XXXX_4$] L 1. Tolerance: • B = ±0.1% • D = ±0.5% • F = ±1% • J = ±5 2. Temperature coefficient of resistance: • E = ±60 ppm/°C (1 MΩ - 7.5 MΩ, 1206 size) • Others = ±50 ppm/°C 3. Taping Reel: • 07 = 7 inch diameter reel • 13 = 13 inch diameter reel 4. Resistance Value: • There are 2~4 digits to indicate the resistor value. Letter R/K/M is decimal point, no need to mention the last zero after R/K/M, e.g.1K2, not 1K20. Detailed resistance rules show in table of “Resistance rule of global part number”.
Characteristics	• Operating Temperature Range: –55 °C to +155 °C • Typical Power Rating: 0.1 W • Maximum Operating Voltage: 75 V • Dielectric Withstand Voltage: 150 V • Maximum Overload Voltage: 150 V

Table 7. Secondary Preference Resistor

Secondary Preference	
Yaego AC Series, 0603 size	
Datasheet	https://www.vishay.com/docs/20035/dcrcwe3.pdf
Resistance Range	• 10 Ω - 1MΩ • Jumper (0Ω)
Part Number Specifications	CRCW 0603 [XXXX1] [X2] [X3] [XX4] 1. Resistance Value: • R = Decimal • K = Thousand • M = Million • 0000 = Jumper 2. Tolerance: • F = ± 1% • J = ± 5% • Z = Jumper 3. Temperature coefficient of resistance: • K = ± 100 ppm/K • N = ± 200 ppm/K • 0 = Jumper 4. Packaging: • EA, EC, ED, EE, EF, EG, EH, EI, EK D11/CRCW 0603 [XXX1] [XXXX2] [X3]% E[XX4] e 1. Temperature coefficient of resistance: • 100 = ± 100 ppm/K • 200 = ± 200 ppm/K 2. Resistance Value: • 10 R = 10 Ω • 10K = 10 KΩ • 1M = 1 MΩ • 0R0 = Jumper 3. Tolerance: • 1 = ± 1% • 5 = ± 5% 4. Packaging: • ET1, ET2, ET3, ET6, ET8, ET9, EF4, E02, E67, E82

Characteristics	• Operating Temperature Range: –55 °C to +155 °C • Typical Power Rating: 0.1 W • Maximum Operating Voltage: 75 V • Permissible Film Temperature: 155 °C • Permissible voltage against ambient (insulation): 100 V

Capacitors

Deviations are highly not recommended.

Table 8. Primary Preference Capacitor

Primary Preference	
Murata GCM Series 0603 Size	
Product Catalogue	Murata Website
Part Number Specifications	GCM 18 8 [XX_1] [XX_2] [XXX_3] [X_4] [XXX_5] [X_6] 1. Temperature Characteristics 2. Rated Voltage 3. Capacitance 4. Capacitance Tolerance 5. Individual Specifications 6. Package

Table 9. Secondary Preference Capacitor

Secondary Preference	
Murata GRT Series 0603 Size	
Product Catalogue	Murata Website
Part Number Specifications	GRT 18 8 [XX_1] [XX_2] [XXX_3] [X_4] [XXX_5] [X_6] 1. Temperature Characteristics 2. Rated Voltage 3. Capacitance 4. Capacitance Tolerance 5. Individual Specifications 6. Package

Connectors

Deviations are not recommended.

General Data

FPC Connector

Table 10. FPC Connector

FPC Connector	
0.5 Pitch | 16 Pin | 505110-1692	
Datasheet	https://www.molex.com/en-us/products/part-detail/5051101692
Specifications	• Max Current: 0.5A per pin • Max Voltage: 50V • Temperature Range: -40°C to 105°C • Pin Count: 16

Connections should look like Figure 1. Keep in mind that connections need to be flipped as same side cables are used.

- First and last 4 pins are power (For a max 2A current)
- Middle 8 pins are signal

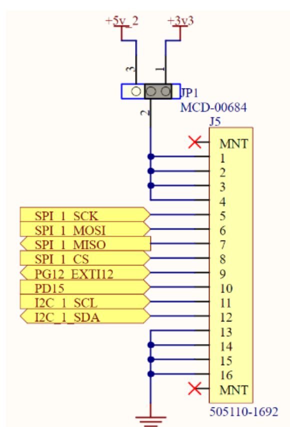

Figure 1. Typical connection of 16 pin FPCs

USB Connectors

Table 11. USB Connector

USB Connector	
USB4105-GF-A-060 Connector	
Design Considerations	• Sits on top of PCB

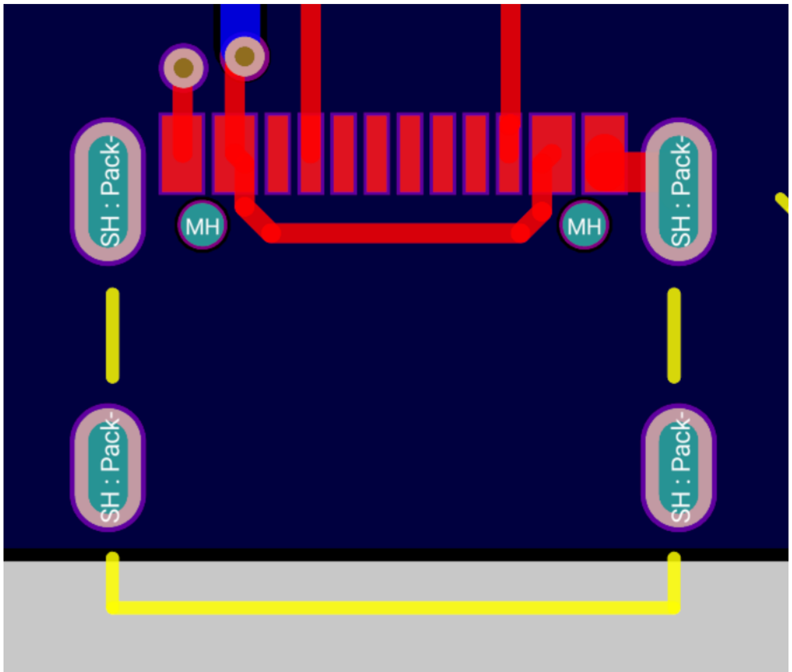

Figure 2. USB4105-GF-A-060 One Sided Surface Mounted USB Connector

Table 12. USB Connector (Type 1)

USB Connector	
UJ20-C-H-G-MSMT-2-P16-TR Connector	
Design Considerations	• Harder to design for PCB • Easier to impedance match correctly • Lower vertical clearance required for PCB

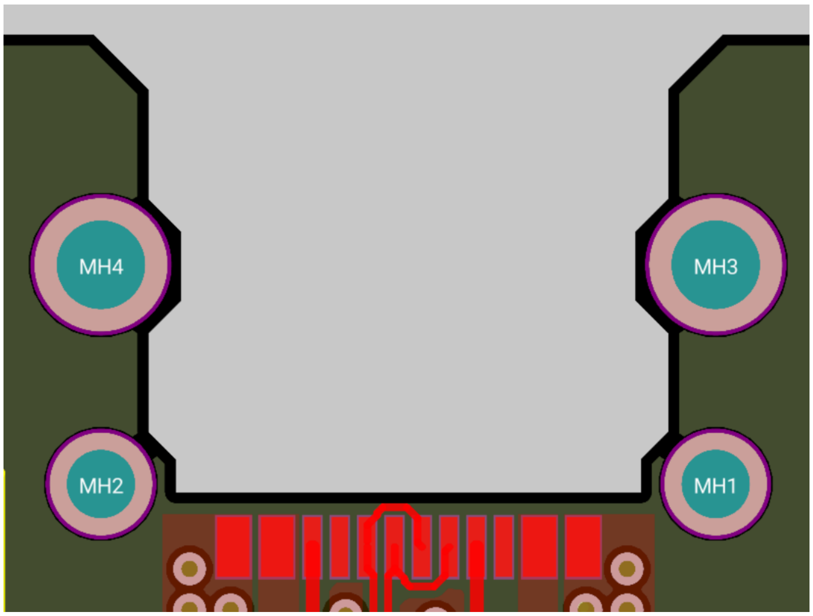

Figure 3. UJ20-C-H-G-MSMT-2-P16-TR Through-PCB USB Connector

0.1 Inch Pin Headers

Table 13. USB Connector (Type 2)

0.1 Inch Pin Headers	
Sources	• Mechatronics Lab (Be mindful of stocks) • Amphol BergStik & EconoStick (Low cost) • Samtech (Low profile): https://samtec.com/solutions/ • Adafruit (Low profile) • OUIIPIN: https://www.altronics.com.au/p/p5430-oupiin-40-way-header-pin/

Dedicated Power

Table 14. Power Headers

Power Headers	
Molex PicoLock 8 Pin	• Cable 15132-0801 • Connector 504050-0891 • For high current subsystems
Molex PicoLock 4 Pin	• Cable 15132-0406 • Connector 504050-0491 • For lower current subsystems

These connectors should be connected with the positive voltage pins in the centre, and the negative voltage pins surrounding the positive voltage pins on the outside

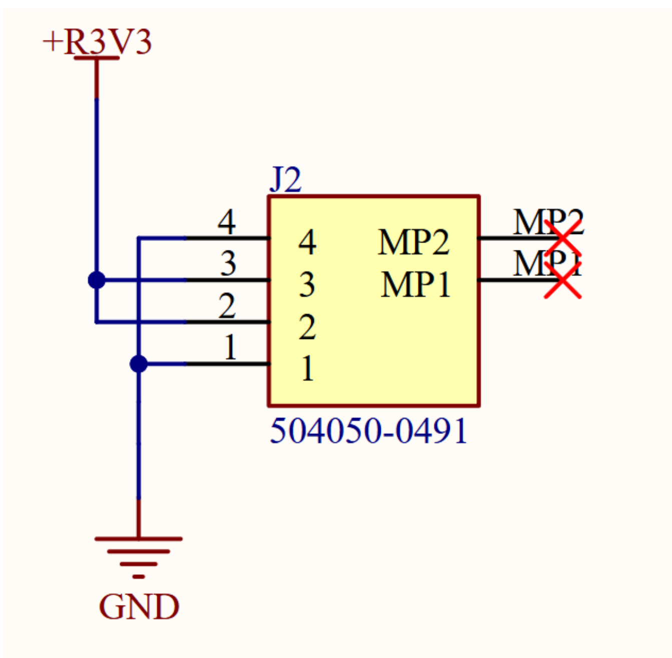

Figure 4. Example Power Connection for the Molex PicoLock 4 Pin

Ferrite Bead

Deviations are encouraged, however must be explained in documentation. Deviations in USB Beads are accepted, however must be explained in documentation.

Table 15. Ferrite Bead

Taiyo Yuden FB series 0603	
Product Catalogue	Mouser Search
Part Number Specifications	FB M [X_1] 1608 [XX_2] [XXX_3] [X_4] T 1. Characteristics: • J = Standard • H = High impedance type 2. Material: • HS • HM • HL 3. Nominal Impedance (eg.) • 330 = 33 Ω • 111 = 110 Ω • 132 = 1300 Ω 4. Impedance Tolerance • - = ± 25 % • N = ± 30 %

USB Power Filtering

Table 16. USB Ferrite Bead (Type 1)

NFZ2MSD181SZ10L	
Maximum DC Current	• 4 A
Impedance	• 180 Ω

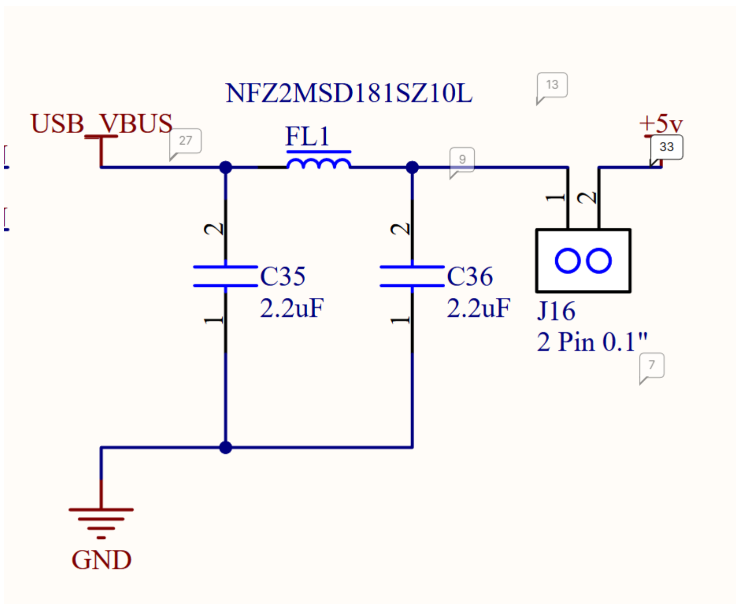

Figure 5. Schematic of USB Power Filtering for NFZ2MSD181SZ10L

Table 17. USB Ferrite Bead (Type 2)

MPZ1608S121ATAH0	
Maximum DC Current	• 2 A
Impedance	• 120 Ω

USB Specifications allow for devices to draw up to 3A, so the MPZ1608S121ATAH0 may be damaged or saturate.

Diodes

Deviations for LEDs are highly not recommended. Deviations for standard diodes are accepted. Deviations for ESD diodes are not recommended.

Table 18. LEDs

Preferred LEDs	
Debug LEDs	• Broadcom 1. HSMH-H190 (Red) 2. HSMG-C190 (Green) 3. HSMA-C190 (Amber)

Table 19. Diodes

Standard Diode	
Information & Use	• BAT60A used for reverse polarity protection • BAT60AE6327HTSA1 Infineon

Table 20. ESD Diodes

Preferred ESD Diodes	
USBLC6-2SC6	STM32 USB ESD – ESD for differential pair & 5V power • Designed for USB 2.0 up to 480 Mb/s • Working voltage: 5.25V • Clamping voltage: 17V • Max current: 5A • Capacitance: 3.5 pF
PESD2USB5UXT-QR	Dual Pin ESD – Good for 2 pin communications • Working voltage: 5V • Clamping voltage: 3.3V • Max current: 4A • Capacitance: 0.6 pF

Crystal Oscillators

Deviations are encouraged, however must be explained in documentation.

Table 21. Crystal Oscillators

Crystal Oscillators	
Product Information	https://ecsxtal.com/products/crystals/surface-mount-crystals/
Useful Information	• Look for low currents & PPM drift from TCXO oscillators

ICs

This section provides a non-exhaustive list of ICs, deviations are accepted, however must be explained in documentation.

Table 22. Current Sensors

Current Sensors	
Types	• INA219 Current & Voltage Sensor • INA236 Current & Voltage Sensor (High accuracy) • INA186 Current Sensor (High accuracy, analogue)

Table 23. MOSFETs

MOSFETS	
Types	• CSD17308Q3 MOSFET

Table 24. STM32s

STM32	
Types	• Low power STM32 microcontrollers (Specific version will need to be researched)

Considerations	• Get larger memory than required (if possible). Sometimes you may need extra memory for larger programs, libraries, or storing data logs

MISC

Table 25. Miscellaneous Components

Miscellaneous Components	
Heat sinking inserts	https://catalog.pemnet.com/category/ultrasonic-heat-staking-inserts
Standoffs	• Pemmet SMD threaded standoff: https://www.pemnet.com/products-overview/products-standoffs/
PTC Resettable Fuse	• PTC RESET FUSE 6V 3.5A 0603 1. Voltage and current ratings can be configured 2. Be aware the fuses are temperature controlled, and may need testing before being used in space

Table 26. Companies

Companies	
Electronics	• Mouser: https://au.mouser.com/
Mechanical	• McMaster Carr: https://www.mcmaster.com